

Introduction to Floquet NESS Engineering with Quantum Optimal Control

Abdrakhman Akchurin

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1 What problem are we solving?

We study a *driven open quantum system* (a qubit) whose state is a density matrix $\rho(t) \in \mathbb{C}^{d \times d}$ ($d = 2$ for a single qubit). The system is:

- **Driven:** we choose time-dependent control functions (e.g. drive amplitude and detuning),
- **Open:** the environment induces relaxation/dephasing, so evolution is not unitary.

The standard Markovian model is the Lindblad master equation

$$\frac{d\rho(t)}{dt} = \mathcal{L}(t)\rho(t) = -i[H(t), \rho(t)] + \sum_k \gamma_k \left(L_k \rho(t) L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho(t)\} \right). \quad (1)$$

This is linear in ρ and generates a completely positive trace-preserving (CPTP) map.

Periodic driving (Floquet setting). Assume $\mathcal{L}(t + T) = \mathcal{L}(t)$ with period T . Define the *one-period propagator* (superoperator)

$$\mathcal{G} \equiv \mathcal{G}(0, T) = \mathcal{T} \exp \left(\int_0^T \mathcal{L}(t) dt \right), \quad \rho(T) = \mathcal{G} \rho(0), \quad (2)$$

where $\mathcal{T}$ is time-ordering. A *Floquet nonequilibrium steady state (NESS)* is a fixed point:

$$\rho_{\text{NESS}} = \mathcal{G} \rho_{\text{NESS}}, \quad \text{Tr}(\rho_{\text{NESS}}) = 1, \quad \rho_{\text{NESS}} \succeq 0. \quad (3)$$

Dissipation typically makes this fixed point unique and attractive (many initial states converge to it under repeated application of $\mathcal{G}$).

Our control goal. We choose the control waveforms inside $H(t)$ (and thus $\mathcal{L}(t)$) to optimize a scalar performance metric evaluated on ρ_{NESS} , e.g. purity $\text{Tr}(\rho_{\text{NESS}}^2)$ or an observable expectation $\text{Tr}(\sigma_\alpha \rho_{\text{NESS}})$.

2 How controls enter the model (concrete parametrization)

For a driven qubit it is common to write a Hamiltonian of the form

$$H(t) = \frac{1}{2} \left(J_x(t) \sigma_x + J_y(t) \sigma_y + \Delta(t) \sigma_z \right), \quad (4)$$

while dissipation is set by Lindblad operators L_k (e.g. relaxation $L = \sigma_-$ and dephasing $L = \sigma_z$) and rates γ_k . You do *not* need the details of L_k to follow the algorithm; what matters is that once $J_x(t), J_y(t), \Delta(t)$ are specified, $\mathcal{L}(t)$ is fully determined.

Finite-dimensional control parameters. To optimize numerically, we represent each control by a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$. Two common choices are:

- **Piecewise-constant (time-slicing):** split a period into N slices and optimize values on each slice.
- **Fourier (smooth, periodic by construction):**

$$J_x(t) = \sum_{m=1}^M \left(a_m \cos \frac{2\pi m t}{T} + b_m \sin \frac{2\pi m t}{T} \right), \quad (5)$$

and similarly for $\Delta(t)$, etc. Then $\boldsymbol{\theta} = \{a_m, b_m, \dots\}$.

3 Numerical core: from $\boldsymbol{\theta}$ to the Floquet map $\mathcal{G}$

Because Eq. (2) involves a time-ordered exponential, we discretize time.

3.1 Normalized time (important for optimizing T)

Introduce normalized time $s = t/T \in [0, 1]$. Then

$$\mathcal{G} = \mathcal{T} \exp \left(\int_0^T \mathcal{L}(t) dt \right) = \mathcal{T} \exp \left(T \int_0^1 \mathcal{L}(s; \boldsymbol{\theta}, T) ds \right). \quad (6)$$

Discretize s on a fixed grid $s_n = (n - \frac{1}{2})/N$ with step $\Delta s = 1/N$. Define the slice Liouvillian

$$\mathcal{L}_n \equiv \mathcal{L}(s_n; \boldsymbol{\theta}, T), \quad \Delta t = T \Delta s = \frac{T}{N}. \quad (7)$$

Then a standard approximation is a product of matrix exponentials:

$$\mathcal{G}(\boldsymbol{\theta}, T) \approx \prod_{n=1}^N \exp(\Delta t \mathcal{L}_n), \quad (8)$$

with the product ordered in time (earlier slices on the right).

3.2 Matrix representation (vectorization)

To compute with linear algebra, represent $\mathcal{L}_n$ as a matrix acting on a vectorized density matrix. Let $\text{vec}(\rho) \in \mathbb{C}^{d^2}$ stack the columns of ρ . Then

$$\text{vec}(\mathcal{L}_n \rho) = \mathbf{L}_n \text{vec}(\rho), \quad \mathbf{L}_n \in \mathbb{C}^{d^2 \times d^2}. \quad (9)$$

Similarly, the one-period map becomes a matrix $\mathbf{G} \in \mathbb{C}^{d^2 \times d^2}$:

$$\text{vec}(\rho(T)) = \mathbf{G} \text{vec}(\rho(0)). \quad (10)$$

In what follows we use bold letters for these matrix forms.

4 Computing the Floquet NESS (fixed point solve)

Let $r \equiv \text{vec}(\rho_{\text{NESS}}) \in \mathbb{C}^{d^2}$. The fixed point equation (3) becomes

$$r = \mathbf{G} r \iff (\mathbf{I} - \mathbf{G}) r = 0. \quad (11)$$

We also require $\text{Tr}(\rho_{\text{NESS}}) = 1$. In vectorized form,

$$\text{Tr}(\rho) = \mathbf{v}^\dagger \text{vec}(\rho), \quad \mathbf{v} \equiv \text{vec}(\mathbb{I}), \quad (12)$$

(up to a transpose convention; $\mathbf{v}^\dagger r = 1$ enforces trace-one).

Practical solve. A robust way is to solve an *augmented linear system*:

$$\begin{bmatrix} \mathbf{I} - \mathbf{G} \\ \mathbf{v}^\dagger \end{bmatrix} r = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (13)$$

This pins down the fixed point by adding the trace constraint. (Positivity $\rho_{\text{NESS}} \succeq 0$ is usually satisfied automatically by CPTP dynamics; if needed, one can add penalties or projection steps.)

5 Objective functions (what we optimize)

Define a scalar objective $\mathcal{J}(\boldsymbol{\theta}, T) = f(\rho_{\text{NESS}}(\boldsymbol{\theta}, T))$. Common examples:

(1) **Purity.**

$$\mathcal{J}_{\text{pur}} = \text{Tr}(\rho_{\text{NESS}}^2). \quad (14)$$

(2) **Expectation value of an observable A (e.g. σ_x, σ_z).**

$$\mathcal{J}_A = \text{Tr}(A \rho_{\text{NESS}}). \quad (15)$$

(3) **Fidelity to a target state $\rho_\star$ (useful for multi-qubit targets).** For example, using the Uhlmann fidelity $F(\rho, \sigma)$.

6 Which controls affect which targets? (control–observable couplings)

For the driven qubit Hamiltonian

$$H(t) = \frac{1}{2} \left(J_x(t) \sigma_x + J_y(t) \sigma_y + \Delta(t) \sigma_z \right), \quad (16)$$

define the Bloch components $r_\alpha(t) \equiv \langle \sigma_\alpha \rangle = \text{Tr}(\sigma_\alpha \rho(t))$ and the instantaneous control vector $\boldsymbol{\Omega}(t) \equiv (J_x(t), J_y(t), \Delta(t))$. Ignoring dissipation momentarily, the unitary part implies

$$\dot{\mathbf{r}}(t) = \boldsymbol{\Omega}(t) \times \mathbf{r}(t), \quad (17)$$

i.e., each control drives a rotation about its axis and therefore couples the *other two* Bloch components. Component-wise, the coherent contributions are

$$\dot{r}_x = J_y r_z - \Delta r_y, \quad (18)$$

$$\dot{r}_y = \Delta r_x - J_x r_z, \quad (19)$$

$$\dot{r}_z = J_x r_y - J_y r_x. \quad (20)$$

Direct coupling map (coherent). Equations (18)–(20) give the key design rule:

- **Targeting $\langle\sigma_x\rangle$:** is driven most directly by J_y (mixes $z \rightarrow x$) and Δ (mixes $y \rightarrow x$).
- **Targeting $\langle\sigma_y\rangle$:** is driven most directly by J_x (mixes $z \rightarrow y$) and Δ (mixes $x \rightarrow y$).
- **Targeting $\langle\sigma_z\rangle$:** is driven most directly by J_x and J_y (mix z with y and x , respectively).
- **Δ does not directly change r_z** in the coherent part; it reshuffles r_x and r_y , which then feed into r_z via J_x, J_y .

Role of dissipation (qualitative but important). The Lindblad terms add damping/attraction on top of the coherent mixing:

- **Dephasing** (e.g. $L = \sigma_z$) suppresses transverse coherences, tending to reduce $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$.
- **Relaxation** (e.g. $L = \sigma_-$) biases the steady state toward a preferred z -polarization; controls change $\langle\sigma_z\rangle$ mainly by continuously mixing population between z and the transverse components.

In Floquet-NESS engineering, the optimized periodic controls exploit this interplay: coherent mixing transfers weight between Bloch components while dissipation stabilizes a unique fixed point.

7 The algorithm: quantum optimal control, stated mathematically

The optimization variables are the control parameters $\boldsymbol{\theta}$ (and possibly T). One iteration of the algorithm is:

1. Given $(\boldsymbol{\theta}, T)$, construct controls $J_x(t), \Delta(t), \dots$ (e.g. Eq. (5)).
2. Build slice Liouvillians $\mathbf{L}_n(\boldsymbol{\theta}, T)$ and step propagators

$$\mathbf{U}_n \equiv \exp(\Delta t \mathbf{L}_n), \quad \Delta t = T/N. \quad (21)$$

3. Form the Floquet map

$$\mathbf{G} = \mathbf{U}_N \mathbf{U}_{N-1} \cdots \mathbf{U}_1. \quad (22)$$

4. Solve for the NESS vector $r = \text{vec}(\rho_{\text{NESS}})$ using Eq. (13).
5. Evaluate the objective $\mathcal{J} = f(\rho_{\text{NESS}})$.
6. Compute gradients $\nabla_{\boldsymbol{\theta}} \mathcal{J}$ (and optionally $\partial \mathcal{J} / \partial T$).
7. Update parameters using a numerical optimizer (e.g. L-BFGS/Adam) with constraints.

The only “nontrivial” step is the gradient computation; the next section shows it explicitly.

8 Gradients: what exactly is differentiated (core math)

We want $\frac{\partial \mathcal{J}}{\partial \theta_k}$ efficiently. There are two dependencies:

$$\theta_k \mapsto \mathbf{L}_n \mapsto \mathbf{U}_n = \exp(\Delta t \mathbf{L}_n) \mapsto \mathbf{G} \mapsto r \mapsto \rho_{\text{NESS}} \mapsto \mathcal{J}.$$

8.1 Derivative of a matrix exponential (Fréchet derivative)

For any matrix function $\mathbf{U}(\theta) = \exp(\mathbf{A}(\theta))$,

$$\frac{d}{d\theta} \exp(\mathbf{A}(\theta)) = \int_0^1 \exp((1-s)\mathbf{A}) \mathbf{A}' \exp(s\mathbf{A}) ds, \quad (23)$$

where $\mathbf{A}' = \frac{d\mathbf{A}}{d\theta}$. In our case, $\mathbf{A} = \Delta t \mathbf{L}_n$, so

$$\frac{\partial \mathbf{U}_n}{\partial \theta_k} = \mathcal{D} \exp(\Delta t \mathbf{L}_n) \left[\Delta t \frac{\partial \mathbf{L}_n}{\partial \theta_k} \right], \quad (24)$$

where $\mathcal{D} \exp[\cdot]$ denotes the Fréchet derivative (often implemented numerically as `expm_frechet`).

8.2 Derivative of the Floquet map $\mathbf{G}$ (product rule)

Because $\mathbf{G} = \mathbf{U}_N \cdots \mathbf{U}_1$,

$$\frac{\partial \mathbf{G}}{\partial \theta_k} = \sum_{n=1}^N \mathbf{U}_N \cdots \mathbf{U}_{n+1} \frac{\partial \mathbf{U}_n}{\partial \theta_k} \mathbf{U}_{n-1} \cdots \mathbf{U}_1. \quad (25)$$

8.3 Derivative of the fixed point r (implicit differentiation)

The fixed point satisfies $(\mathbf{I} - \mathbf{G})r = 0$ and $\mathbf{v}^\dagger r = 1$. Differentiate $(\mathbf{I} - \mathbf{G})r = 0$:

$$(\mathbf{I} - \mathbf{G}) \frac{\partial r}{\partial \theta_k} = \left(\frac{\partial \mathbf{G}}{\partial \theta_k} \right) r. \quad (26)$$

Differentiate the trace constraint $\mathbf{v}^\dagger r = 1$:

$$\mathbf{v}^\dagger \frac{\partial r}{\partial \theta_k} = 0. \quad (27)$$

Together, Eqs. (26)–(27) form an augmented linear system for $\partial r / \partial \theta_k$:

$$\begin{bmatrix} \mathbf{I} - \mathbf{G} \\ \mathbf{v}^\dagger \end{bmatrix} \frac{\partial r}{\partial \theta_k} = \begin{bmatrix} \left(\frac{\partial \mathbf{G}}{\partial \theta_k} \right) r \\ 0 \end{bmatrix}. \quad (28)$$

8.4 Derivative of the objective

If $\mathcal{J} = f(\rho_{\text{NESS}})$ depends on parameters only through r , then by the chain rule

$$\frac{\partial \mathcal{J}}{\partial \theta_k} = \left(\frac{\partial \mathcal{J}}{\partial r} \right) \cdot \frac{\partial r}{\partial \theta_k}. \quad (29)$$

For example:

- If $\mathcal{J}_A = \text{Tr}(A \rho_{\text{NESS}})$, then $\frac{\partial \mathcal{J}}{\partial \rho_{\text{NESS}}} = A$ and the corresponding vector gradient is a reshaped version of A .
- If $\mathcal{J}_{\text{pur}} = \text{Tr}(\rho_{\text{NESS}}^2)$, then $\frac{\partial \mathcal{J}}{\partial \rho_{\text{NESS}}} = 2\rho_{\text{NESS}}$.

Computational note. Eq. (28) requires solving one linear system per parameter θ_k . For many parameters, an *adjoint method* can reduce cost, but Eq. (28) is the cleanest way to explain what is happening mathematically.

9 Selecting the period via a T -scan (outer loop)

Rather than treating the period T as a continuous optimization variable (which introduces additional sources of numerical non-smoothness through $\Delta t = T/N$ and through the T -dependence inside the controls), we adopt a simple and robust *outer-loop scan* over T .

Idea. Fix a physically meaningful interval $T \in [T_{\min}, T_{\max}]$ and evaluate the best achievable NESS objective at each candidate period by optimizing only waveform parameters θ :

$$\mathcal{J}^*(T) \equiv \max_{\theta \in \Theta} \mathcal{J}(\theta, T). \quad (30)$$

Then select

$$T^* \in \arg \max_{T \in \mathcal{T}_{\text{grid}}} \mathcal{J}^*(T), \quad (31)$$

where $\mathcal{T}_{\text{grid}} = \{T_1, \dots, T_K\}$ is a discrete grid.

Implementation (algorithmic recipe). For each $T_k \in \mathcal{T}_{\text{grid}}$:

1. Fix $T = T_k$ and discretize normalized time $s \in [0, 1]$ with a fixed N (so $\Delta t = T_k/N$).
2. Optimize θ (Fourier coefficients or time-slice amplitudes) to maximize $\mathcal{J}(\theta, T_k)$.
3. Record $\mathcal{J}^*(T_k)$ and the corresponding optimal controls.

Optionally, perform a second, refined scan around the best coarse-grid period.

Why this is paper-friendly. A T -scan cleanly separates (i) *physics*—how the Floquet period changes the steady-state structure—from (ii) *waveform design* at fixed T . It also makes it straightforward to present a diagnostic plot $\mathcal{J}^*(T)$ versus T and to justify the chosen T^* .

Choosing the scan range (one practical constraint). For Fourier controls truncated at harmonic index M , the highest angular frequency scales as $\omega_{\max} \sim 2\pi M/T$. Hardware or modeling bandwidth limits therefore imply a lower bound on T :

$$T \geq T_{\min} \approx \frac{2\pi M}{\omega_{\max, \text{phys}}}. \quad (32)$$

This prevents artificially fast modulation that would be unphysical in an experimental implementation.